Jungsoo Kim

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Research Interests

- Uncertainty quantification and reliability in AI; conformal prediction and interpretable uncertainty; causal representation learning under unobserved confounding; reliability of AI-enhanced imagery. Applications span healthcare, policy evaluation, and evidentiary use.

Education

Ph.D. **Seoul National University, Graduate School of Data Science**, Data Science — Seoul, Korea
- Advisor: Sanghack Lee; Causality Lab. — Sept 2024 – present

M.S. **Seoul National University, Graduate School of Data Science**, Data Science — Seoul, Korea
- GPA: 4.22/4.5 — Mar 2022 – Aug 2024
- Course Project: Implemented and parallelized a GRU-based name generation model from scratch using C/C++ and CUDA as part of a high-performance computing course.

B.P.A. / B.B. **Yonsei University, Underwood International College**, Science, Technology and Policy; Quantitative Risk Management — Seoul, Korea
- GPA: 4.15/4.5 — Mar 2017 – Feb 2022

Exchange Student **National University of Singapore, Yale-NUS College**, Mathematical and Computational Science — Singapore, Sept 2019 – Feb 2020

Selected Publications

Decomposing Conformal Uncertainty: Calibration- and Instance-Driven Feature Attribution — Jan 2026
Accepted as a poster. Equal contribution (co-first authors): Sangyeon Cho, Minyoung Cho, and Jungsoo Kim.
Sangyeon Cho, Minyoung Cho, *Jungsoo Kim*, Sujeong Oh, Sanghack Lee
(Advances in Neural Information Processing Systems (NeurIPS))

A Tale of Two Uncertainties: Global-Local Attribution for Conformal Prediction — Jan 2026
Spotlight paper; earlier workshop version of the NeurIPS 2026 paper. Equal contribution (co-first authors): Sangyeon Cho, Minyoung Cho, and Jungsoo Kim.
Sangyeon Cho, Minyoung Cho, *Jungsoo Kim*, Sujeong Oh, Sanghack Lee
openreview.net/forum?id=2aIvedJMpp (ICML 2026 Workshop on Epistemic Intelligence in Machine Learning (EIML))

Instrumental Variable Representation Learning under Confounded Covariates — Jan 2025
Jungsoo Kim, Kwonho Kim, Inwoo Hwang, Sanghack Lee
(NeurIPS 2025 Workshop on CauScien: Uncovering Causality in Science)

When Mass Atrocities Come to Light in the Artificial Eyes in the Sky: AI and the Use of Satellite Images for International Criminal Accountability
Revised and resubmitted; under review.
Jungsoo Kim, Heejin Kim, Sanghack Lee
(Humanities and Social Sciences Communications (Springer Nature))

Selected Research Projects

Conformal uncertainty and patient-specific retesting

- Originated the motivating use case and led experimental design: constructed semi-synthetic data from real sepsis data and designed a risk score with patient-specific retesting needs.
- Compared retest-variable selection using total attribution, equal calibration/instance weighting, and greater instance weighting; showed that emphasizing the test instance better prioritized patient-specific retest variables under a limited testing budget in simulation.

Instrumental variable representation learning under confounded covariates

- Conceived the research question and designed an architecture to learn IV representations from high-dimensional covariates correlated with unobserved confounders, targeting causal effect estimation under weaker assumptions.
- Implemented the model, ran the experiments, and wrote the manuscript, carrying the project from initial concept through empirical evaluation.

Reliability of AI-enhanced satellite imagery as evidence

- Investigated the reliability of AI-enhanced satellite imagery for international criminal accountability; ran model inference across 15 super-resolution setups and conducted quantitative and qualitative analyses.
- Combined quantitative and qualitative evidence of reliability limitations with analysis of international criminal accountability, highlighting risks of the evaluated AI-enhanced imagery as evidence; manuscript under review.

Research and Industrial Experience

Ministry of Food and Drug Safety (MFDS) Project, Project Lead for Causal Inference Section — Mar 2025 – Dec 2025

- Led the causal inference section of the MFDS project, applying conformal prediction to analyses of statin effects on blood glucose.
- Investigated variation in prediction interval widths across individuals; this practical problem motivated subsequent research on feature attribution for conformal uncertainty.

SK Center for Social Value Enhancement Studies, Research Assistant — Aug 2021 – Jan 2022

- Analyzed beneficiary financial records totaling KRW 10 billion, estimated the following fiscal year's budget, and built a research database on the Korean social economy.

Mooncademy (Korean Traditional Music Education Start-up), Product Manager — Oct 2020 – Apr 2021

- Planned website information architecture with engineers and designers; launched a gayageum lecture crowdfunding campaign raising KRW 10,491,700.

Teaching and Service

Teaching Assistant, Seoul National University
Causal Inference; Machine Learning and Deep Learning II; Basic Statistics & Mathematics Bootcamp.

Python Instructor, SNU Language Education Institute Outreach — 2025-07 to 2025-08
Taught Python and basic algorithms to Malaysian high school students, emphasizing collaborative problem-solving.

Selected Honors and Writing

5th POSTECH SF Award, Excellence Prize — Jan 2025
Authored "Will of Stochastic Ghost," published in Diagonal Argument (Eunhengnamu, 2025); fiction exploring generative AI and evidence.

Journal of Korea Artificial Intelligence Association, Best Paper Jan 2024
Submitted a master's project paper on instrumental variables.

CSES Social Value Research Idea Award, Excellence Prize Jan 2024
Proposed a study on causal structure analysis for Korean population policy.

LH COMPAS Land & City Data Competition, Excellence Prize Jan 2024
Identified medically underserved areas in Asan through geospatial analysis.

EdTech Data Analysis Hackathon, Grand Prize Jan 2023
Built a dropout-prediction model using survival analysis.

Public BigData Analysis Hackathon, Excellence Prize Jan 2020
Developed a traffic-accident risk model with XGBoost and interpreted drivers using SHAP.

Skills and Languages

Programming: Python, R, C++, C, SQL; CUDA C/C++ (coursework)

Research Methods: Conformal prediction and uncertainty quantification; instrumental variables and causal representation learning; experimental design; quantitative and qualitative model evaluation

ML and Geospatial Tools: PyTorch; super-resolution and diffusion model workflows (SRCNN, SRGAN, StableSR, Stable Diffusion); GeoPandas, QGIS

Data Systems: Neo4j, BigQuery; Kafka (event streaming)

Languages: Korean, English (TOEFL 100; OPIc AL)